

Equal Rank and Equal Spectral Mass Do Not Determine the First-Alias Weighted Prefix

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Abstract

We construct two finite normal spectral models on the physical first-alias clock that agree in a stringent package of coarse invariants compatible with the noisy Hardy-head ceilings but have opposite strict-prefix behavior. Let $\mathcal{U}_L(r)$ be the complete L th-root shell of radius r , let $\mathcal{Y}_k$ be the RH-342 counterloop, and set

$$a = \frac{3}{4}, \quad b = \frac{4}{5}, \quad c = \sqrt{\frac{481}{800}}.$$

The multisets

$$\mathcal{X}_k^{\text{inv}} = \mathcal{Y}_k \sqcup \mathcal{U}_{4k}(c), \quad \mathcal{X}_k^{\text{vis}} = \mathcal{Y}_k \sqcup \mathcal{U}_{2k}(a) \sqcup \mathcal{U}_{2k}(b)$$

both have rank $6k - 2$ and squared spectral mass $(2k - 2)\beta_k^2 + 481k/200$. They are conjugation closed, eventually simple, normally realizable, contained in the same Hardy annulus, and have the same maximum modulus β_k . Both agree with $\mathcal{Y}_k$ at every order $2 \leq n < 2k$, hence at every fixed order eventually, but they split at the first alias $n = 2k$.

For the RH-299 budget

$$D_m(X, Y) = \sum_{2 \leq n < m} \frac{|p_n(X) - p_n(Y)| R^n}{n}, \quad R = \frac{7}{5},$$

the strict endpoint is decisive:

$$D_{4k}(\mathcal{X}_k^{\text{inv}}, \mathcal{Y}_k) = 0, \quad D_{4k}(\mathcal{X}_k^{\text{vis}}, \mathcal{Y}_k) = \left(\frac{21}{20}\right)^{2k} + \left(\frac{28}{25}\right)^{2k} \rightarrow \infty.$$

The genus-one quotient factors are respectively $1 - (cz)^{4k}$ and $[1 - (az)^{2k}][1 - (bz)^{2k}]$. Thus equal rank, squared mass, cap, maximum modulus, normal simplicity, and all pre-alias or eventual fixed-order data do not determine the moving strict-prefix budget. This is an exact finite spectral information-class theorem, not an actual noisy-operator theorem. No physical head transport, determinant gluing, Gate progress, or Riemann-hypothesis claim follows.

1 Source-locked counterloop and complete shells

Fix the common Hardy constants and physical clock

$$q = \frac{1}{2}, \quad r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad u = 4k. \quad (1)$$

RH-342 locks the graded counterloop in this normalization as

$$\mathcal{Y}_k = \{\beta_k e^{ij\pi/k}, \beta_k e^{-ij\pi/k} : 1 \leq j \leq k - 1\}, \quad \#\mathcal{Y}_k = 2k - 2, \quad (2)$$

with exact moments

$$p_n(\mathcal{Y}_k) = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n). \quad (3)$$

Here and below $p_n(X) = \sum_{x \in X} x^n$. The radius obeys

$$\beta_k \longrightarrow \beta = (r_H \sqrt{\lambda})^{-1} = 0.9080523604 \dots \quad (4)$$

by RH-272, and $\beta_k < 1/r_H$ eventually [2, 4]. The strict prefix $2 \leq n < 4k$ contains $2k$ and excludes $4k$.

For $L \geq 2$ and $r > 0$, define the complete root shell

$$\mathcal{U}_L(r) = \{r e^{2\pi i j/L} : 0 \leq j < L\}. \quad (5)$$

Proposition 1.1 (Complete-shell power ledger). *For every integer $n \geq 1$,*

$$p_n(\mathcal{U}_L(r)) = L r^n \mathbf{1}_{L|n}. \quad (6)$$

Proof. After factoring out r^n , the remaining sum is the finite geometric progression

$$\sum_{j=0}^{L-1} e^{2\pi i j n/L},$$

which equals L when $L \mid n$ and zero otherwise. This is the complete shell identity used in RH-289 [6]. $\square$

Put

$$a = \frac{3}{4}, \quad b = \frac{4}{5}, \quad c = \sqrt{\frac{481}{800}}, \quad c^2 = \frac{a^2 + b^2}{2}, \quad (7)$$

and define two model spectra

$$\mathcal{X}_k^{\text{inv}} = \mathcal{Y}_k \sqcup \mathcal{U}_{4k}(c), \quad \mathcal{X}_k^{\text{vis}} = \mathcal{Y}_k \sqcup \mathcal{U}_{2k}(a) \sqcup \mathcal{U}_{2k}(b). \quad (8)$$

The superscripts describe visibility before the strict endpoint. Neither multiset is named or interpreted as the actual spectral head of K_σ/r_H .

2 Equal rank, mass, cap, and normal realizability

We use *squared spectral mass* for the multiset quantity

$$M_2(X) = \sum_{x \in X} |x|^2. \quad (9)$$

For a normal diagonal realization this also equals its squared Hilbert–Schmidt norm; no such operator-language identification is used for an actual noisy operator.

Theorem 2.1 (Exact equal-invariant package). *For every $k \geq 2$,*

$$\#\mathcal{X}_k^{\text{inv}} = \#\mathcal{X}_k^{\text{vis}} = 6k - 2, \quad (10)$$

$$M_2(\mathcal{X}_k^{\text{inv}}) = M_2(\mathcal{X}_k^{\text{vis}}) = (2k - 2)\beta_k^2 + \frac{481k}{200}. \quad (11)$$

Both multisets are closed under complex conjugation and are spectra of finite normal diagonal matrices. For all sufficiently large k , they are simple, every root lies in

$$q < |z| < 1/r_H, \quad (12)$$

and both have maximum root modulus exactly β_k .

Proof. The added ranks are $4k$ on both sides, so (10) follows from $\#\mathcal{Y}_k = 2k - 2$. The invisible shell adds squared mass

$$4kc^2 = 4k \frac{481}{800} = \frac{481k}{200}.$$

The visible shells add

$$2ka^2 + 2kb^2 = 2k \left(\frac{9}{16} + \frac{16}{25} \right) = 2k \frac{481}{400} = \frac{481k}{200},$$

which proves (11).

Every complete shell and $\mathcal{Y}_k$ is conjugation closed, hence so are their disjoint unions. Listing the roots on the diagonal gives finite normal realizations. The exact squared comparisons

$$a^2 = \frac{450}{800} < c^2 = \frac{481}{800} < b^2 = \frac{512}{800} \quad (13)$$

give $a < c < b$. Since $q < a$ and the limit in (4) is larger than $b = 0.8$, eventually

$$q < a < c < b < \beta_k < 1/r_H. \quad (14)$$

Each individual root shell is simple. Their radii are then pairwise distinct, while the roots in $\mathcal{Y}_k$ are the distinct nonreal $2k$ th roots on radius β_k . Thus both unions are simple. Equation (14) also proves (12) and shows that the common maximum modulus is β_k . $\square$

Corollary 2.2 (Compatibility with only the coarse physical ceilings). *On the clock (1), the rank and squared spectral mass of both candidates satisfy*

$$O(k) = O(\log(1/\sigma)) = o(1/\sigma). \quad (15)$$

Consequently they do not violate the RH-282 ceilings

$$r_\sigma \leq 4/\sigma, \quad M_\sigma \leq 1/\sigma, \quad (16)$$

but those inequalities do not physically realize either candidate.

Proof. The rank formula is linear in k . Since β_k converges, the mass in (11) is also $O(k)$. Equation (1) and the elementary relation $\log(1/\sigma) = o(1/\sigma)$ give (15). RH-282 proves (16) for the actual noisy spectral data [7]; satisfying upper ceilings is only a consistency check, not an identification theorem. $\square$

3 Pre-alias agreement and the first split

Theorem 3.1 (Exact moving-order moment dichotomy). *For every $n \geq 1$,*

$$p_n(\mathcal{X}_k^{\text{inv}}) - p_n(\mathcal{Y}_k) = 4kc^n \mathbf{1}_{4k|n}, \quad (17)$$

$$p_n(\mathcal{X}_k^{\text{vis}}) - p_n(\mathcal{Y}_k) = 2k(a^n + b^n) \mathbf{1}_{2k|n}. \quad (18)$$

Hence both candidates agree exactly with $\mathcal{Y}_k$ for every

$$2 \leq n < 2k. \quad (19)$$

For every fixed $n \geq 2$, this agreement therefore holds for all sufficiently large k . The candidates do not both agree with $\mathcal{Y}_k$ throughout $2 \leq n < 4k$: at $n = 2k$,

$$p_{2k}(\mathcal{X}_k^{\text{inv}}) - p_{2k}(\mathcal{Y}_k) = 0, \quad p_{2k}(\mathcal{X}_k^{\text{vis}}) - p_{2k}(\mathcal{Y}_k) = 2k(a^{2k} + b^{2k}) > 0. \quad (20)$$

Proof. Apply Proposition 1.1 to each shell in (8). No positive integer $n < 2k$ is divisible by either $2k$ or $4k$, proving (19). For fixed n , eventually $n < 2k$. Finally, $2k$ is not divisible by $4k$ but is divisible by $2k$, which gives (20). $\square$

The distinction in Theorem 3.1 is essential. The invisible candidate agrees with $\mathcal{Y}_k$ at every order below $4k$, while the visible candidate first differs at $2k$. Thus the theorem does not claim that the two candidates carry identical data on the whole strict prefix.

4 Exact strict-prefix budgets

Following RH-299, define for finite multisets X, Y and $m \geq 3$

$$D_m(X, Y) = \sum_{n=2}^{m-1} \frac{|p_n(X) - p_n(Y)| R^n}{n}, \quad R = \frac{7}{5}. \quad (21)$$

The denominator n is part of the determinant-logarithm normalization and must not be dropped [5].

Theorem 4.1 (Zero versus divergent first-alias budget). *For every $k \geq 2$,*

$$D_{4k}(\mathcal{X}_k^{\text{inv}}, \mathcal{Y}_k) = 0, \quad (22)$$

$$D_{4k}(\mathcal{X}_k^{\text{vis}}, \mathcal{Y}_k) = \left(\frac{21}{20}\right)^{2k} + \left(\frac{28}{25}\right)^{2k}. \quad (23)$$

In particular, the second budget tends to infinity as $k \rightarrow \infty$.

Proof. There is no multiple of $4k$ in $2 \leq n < 4k$, so (17) makes every summand in (21) zero, proving (22). The only multiple of $2k$ in the same strict prefix is $n = 2k$. Therefore (18) gives exactly

$$\begin{aligned} D_{4k}(\mathcal{X}_k^{\text{vis}}, \mathcal{Y}_k) &= \frac{2k(a^{2k} + b^{2k})R^{2k}}{2k} \\ &= (aR)^{2k} + (bR)^{2k} \\ &= \left(\frac{21}{20}\right)^{2k} + \left(\frac{28}{25}\right)^{2k}. \end{aligned}$$

This retains the exact cancellation between the shell multiplicity $2k$ and the logarithmic denominator $n = 2k$. Both bases are larger than one, proving divergence. $\square$

Remark 4.2 (Why the endpoint must remain strict). The first nonzero moment of $\mathcal{U}_{4k}(c)$ occurs at $n = 4k$. Thus replacing $n < 4k$ by $n \leq 4k$ destroys (22). The paper proves a strict-prefix theorem only.

5 Exact genus-one quotient factors

For a finite multiset X , put

$$E_1(w) = (1 - w)e^w, \quad \Phi_X(z) = \prod_{x \in X} E_1(xz). \quad (24)$$

Proposition 5.1 (Candidate quotient factors). *The shell quotients are the exact entire factors*

$$\frac{\Phi_{\mathcal{X}_k^{\text{inv}}}(z)}{\Phi_{\mathcal{Y}_k}(z)} = 1 - (cz)^{4k}, \quad (25)$$

$$\frac{\Phi_{\mathcal{X}_k^{\text{vis}}}(z)}{\Phi_{\mathcal{Y}_k}(z)} = [1 - (az)^{2k}][1 - (bz)^{2k}]. \quad (26)$$

Proof. For every complete shell with $L \geq 2$, its first power sum is zero, so all linear exponential terms in (24) cancel. The elementary root factorization then gives

$$\prod_{u \in \mathcal{U}_L(r)} E_1(uz) = 1 - (rz)^L. \quad (27)$$

Since each candidate is the disjoint union of $\mathcal{Y}_k$ and its displayed shells, equations (25) and (26) follow. $\square$

These are finite analytic factor identities. They do not identify either quotient with a factor in one physical noisy determinant.

6 Sharp information-class underdetermination

Corollary 6.1 (Equal coarse data do not determine the moving budget). *Within the class of finite complex spectral multisets, the following data, even when imposed simultaneously, do not determine the moving strict-prefix quantity $D_{4k}(X, \mathcal{Y}_k)$:*

1. rank $6k - 2$ and squared spectral mass $(2k - 2)\beta_k^2 + 481k/200$;
2. the common cap $1/r_H$ and common maximum modulus β_k ;
3. simple, conjugation-closed finite normal realizability;
4. equality with $\mathcal{Y}_k$ at every pre-alias order $2 \leq n < 2k$; and
5. equality with $\mathcal{Y}_k$ at every fixed order eventually.

Indeed, the same data allow the exact value zero and a value tending to infinity. A future actual rank cap $r_\sigma \leq 2k - 2$ would exclude both examples.

Proof. Theorem 2.1 supplies items 1–3 for both candidates, and Theorem 3.1 supplies items 4–5. The two values of the moving budget are given by Theorem 4.1. Both candidates have rank $6k - 2 > 2k - 2$, proving the final exclusion statement. $\square$

Corollary 6.1 is sharp in its declared information class: it preserves not only rank and mass but also the maximum radius and all pre-alias data. It does not refute a future transport theorem that uses actual operator structure, a sharper rank cap, root matching, moving-order moments, contour information, or an annular norm.

RH-303 proves that actual annular convergence would force fixed-order head transport [3]. Our examples show only that, in a finite normal information class, eventual fixed-order transport is not by itself sufficient for the moving $4k$ budget. No model-to-actual substitution is made.

7 Protocol, verdict, and claim firewall

The executable artifact evaluates the exact shell indicators, common ranks, the rational squared-mass identity, the first split, both strict-prefix budgets, and the radius ordering. Its finite rows at $k = 3, 5, 9, 17$ reproduce closed identities only. They are not numerical fits, noisy eigenvalue observations, or asymptotic evidence.

On the physical clock, the examples are compatible with only the coarse RH-282 rank and mass ceilings. Such compatibility cannot identify them with the actual noisy head. Accordingly the exact route verdict is:

1. equal rank, equal squared mass, common cap, and fixed-order data are **STOP_SCOPED** as an inference to the moving prefix without a sharper actual rank law or moving-order physical input;
2. actual alias-inclusive head transport remains **NOT_TESTABLE/open**; and
3. the RH-288 determinant-gluing leaf remains inactive.

In particular, this paper proves neither divergence nor vanishing of the actual D_{4k} , no physical rank mismatch, no equivalence of the physical p and q prefixes, and no failure of all future transport mechanisms. It does not activate RH-288, identify the model spectra with K_σ , or prove determinant gluing. Gates A–E remain false/open as reviewed in RH-341 [1]. No Hilbert–Polya operator is constructed, no Riemann zero is identified, no von Mangoldt trace formula or completed-zeta divisor equality is proved, and the Riemann Hypothesis is not proved.

References

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